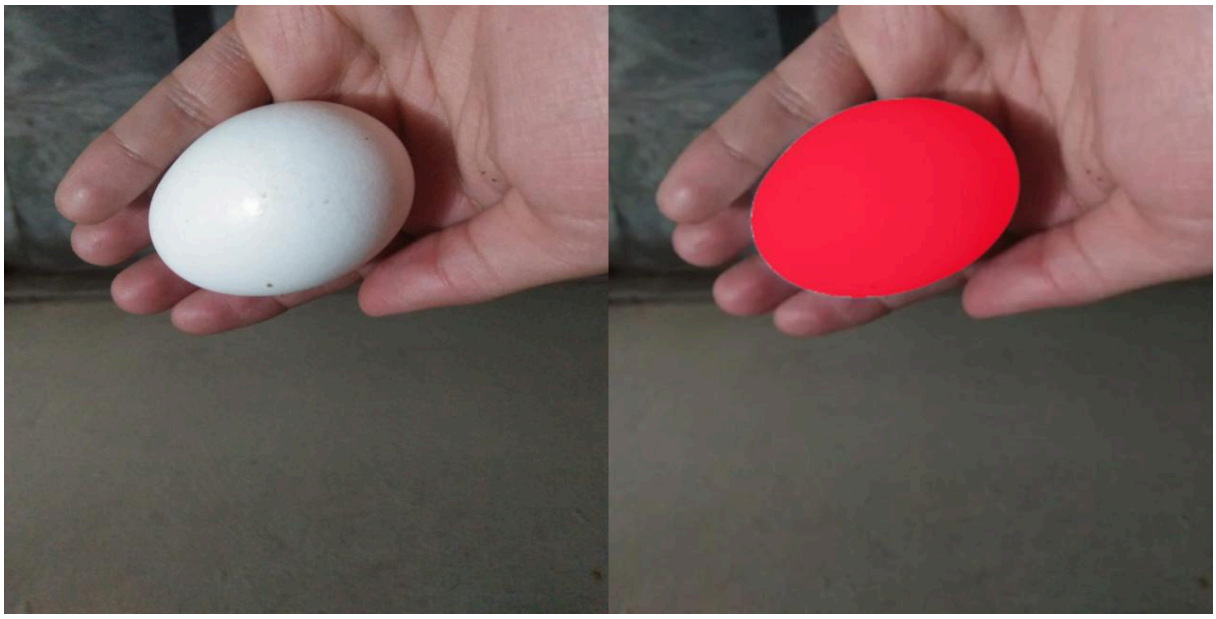
CV Intern Task

Dataset: Roboflow Link | Alternative zip link

Task 1 - Segmentation using Pretrained Model

- Use a pretrained segmentation model (for example, SAM) to identify and segment the egg(s) in all images within the dataset.
- Example



Task 2 - Segmentation using Image Processing

- Split the dataset into 2 parts, a dev split and an eval split, in a 4:1 ratio. M**ake sure to reduce data leakage between the development and eval splits.**
- Develop a segmentation algorithm using classical image processing techniques to segment eggs in the images in the dataset. Try to closely match the segmentation quality produced in Task 1.
- Provide an experiment log in the report covering **trials and improvements that you did, including unsuccessful attempts**. For each trial, document what changed, why you tried it, the parameters used, the development-set results, and why you kept or discarded the change. Show how these experiments led to your algorithm.

Task 3 - Evaluation and Analysis

- Considering Task 1 segmentations as ground truth and the image-processing-based method's outputs as predictions, compute relevant segmentation evaluation metrics and produce a detailed analysis report highlighting performance gaps and potential areas for improvement. This ideally should include both quantitative and **qualitative evaluation**. Qualitative evaluation should include a few visualizations of wrongly segmented samples along with other plots highlighting what went wrong in your algorithm with a detailed but brief writeup.

Task 4 - Video Tracking

- Run your final algorithm on this YouTube video. The objective is to track individual eggs separately.
- You are open to modifying your egg segmentation algorithm and/or implementing a tracking algorithm.

AI tools

You can use AI coding tools. Briefly disclose where you've used them. You're not expected to outsource your thinking to these tools and are expected to clearly understand and explain the underlying methods used in these tasks. The initial interview will involve you walking through your algorithm implementation, adding modifications as suggested by your interviewer as expected.

Submission

- Create a publicly accessible GitHub repo for this, along with a PDF report of the details asked for in all of the above tasks. Make sure to add that PDF report to the GitHub repo. That PDF report should have a link to a publicly accessible Google Drive link to the video tracking task output.
- Submit your work within 7 days of receiving the assignment. Partial submissions are acceptable.
- Make sure to include instructions to set up the repo and run the experiments.